Investigator Working Notes

Tanner Smith

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Investigator Working Notes on **Evidence Item**: **001** (**Lexar USB Drive**)

Case: 2026-05-12-CORP (Insider Threat Investigation)

Investigator: Tanner Smith

Date: May 12, 2026

Collection/Preservation

• **Day 1** - 14:00: HR delivered a Lexar USB flash drive (image5.dd.001) discovered in a terminated employee's desk. Initiated Chain of Custody (CoC) to ensure legal standing. According to NIST (2022) guidelines, maintaining a continuous record of contact is vital for admissibility. Connected via hardware write-blocker to prevent metadata alteration.

- Evidence Tag: Item 001
- Original Evidence: Lexar USB Flash Drive
- Image File: image5.dd.001
- Size: 3824 MB
- MD5 Hash: 51d3851fe4181ceb98fff16e11cc6e71

• **Day 2** - 09:00: HR representative and local IT support returned. IT provided a new forensic image of the corporate system hard drive belonging to the suspect. Gently advised HR that this must be treated as a new artifact and processed securely before merging with the overall investigation. 09:15: Updated CoC and Evidence Control Log. Uniquely tagged the new evidence as "winimage.001". Initiated hashing procedures to create a forensically sound working copy.

- Evidence Tag: Item 002
- Original Evidence: VMware Virtual IDE Hard Drive
- Image File: winimage.001
- Size: 15360 MB
- MD5 Hash: d39c651a4f522d2e13bd0429e2de9e3f

• **Day 3 -** 10:00: Initiated Chain of Custody (CoC) and Evidence Control Log (ECL) for the newly received evidence to maintain legal admissibility per NIST (2006) guidelines. 10:15: The evidence item was uniquely identified and tagged as Item 003 (System Memory Capture File). 10:45: To preserve the original evidence, a forensic copy was created. The original file was hashed to verify integrity and establish a baseline before securely transferring the working copy to the forensic workstation for upcoming volatile memory analysis. 11:00: The original evidence was secured in Evidence Locker #1. The working copy was staged for subsequent processing utilizing the Volatility memory forensics framework.

- Evidence Tag: Item 003
- Original Evidence Name: Windows 10 x64-31acf216.vmem
- Size: 2,097,153 KB (2 GB)
- MD5 Hash: 33201c5f2cfd81bbcafbbbd371122431

- SHA-256 Hash: fad7c7afbedcd00680f3dd5b43afe09ec32ab76541be9562c1836b33a41cf2d4
- CRC32 - b09e7deb

Analysis in Autopsy

Item 1: USB Drive - image5.dd.001

- 15:00: Loaded image5.dd.001 into Autopsy v4.21.0.
- 15:30: Located four key PDF documents establishing suspect identity as Nicholas Anderson (N.A.) and motive (disgruntled over transfer/performance).
- 16:00: Evaluated stuff.txt. Autopsy flagged for suspected encryption (entropy 7.999982).
- 16:20: Identified VeraCrypt Setup 1.26.7.exe in the root directory, providing the methodology used to encrypt stuff.txt.

Item 2: System Drive - winimage.001

- 10:00: Added winimage.001 as a new data source to the existing Autopsy case. Initiated ingest modules.
- 10:30: Conducted manual file system navigation while ingest ran.
- 10:45: Item found: C:\Program Files\VeraCrypt and C:\Users\Nick\Documents\VeraCrypt. Relevance: Confirms the suspect successfully installed the encryption tool on the corporate machine.

- 11:00: Item found: C:\Users\Nick\Documents\1.txt. Relevance: File size is exactly 10MB and shows high entropy. Suspected secondary encrypted container, mimicking the behavior seen with stuff.txt on the USB drive.
- 11:15: Item found: SYSTEM Registry Hive (MountedDevices). Relevance: Devastating finding. Proves that VeraCrypt volumes were actively mounted to the corporate system as drives K: and L:.
- 11:30: Item found: Windows Recent LNK Files (Local Disk (K).lnk, Local Disk (L).lnk, 1.lnk). Relevance: Places the user (Nick) actively navigating inside the encrypted volumes. 1.lnk points to a file accessed within the encrypted L: volume.
- 11:45: Item found: C:\Users\Nick\Documents\7z2201-x64.exe. Relevance: 7-Zip installer. Potential staging tool used to compress company databases before encrypting them.
- 12:00: Item found: C:\Users\Nick\Documents\Imager_Lite_3.1.1. Relevance: Identifies FTK Imager Lite, the forensic tool used by IT personnel prior to evidence handover. Highly likely to be the source of the missing RAM capture mentioned by HR.

Item 3 Volatile Memory - Windows 10 x64-31acf216.vmem

- 13:00: Loaded the memory image into the Volatility command-line framework and ran initial image identification modules to determine the operating system profile.
- 13:30: Executed the process-listing plugins (“pslist” & “pstree”) to map out all active system and user processes running at the time of the capture.
- 13:45:

Investigative Scenario Updates

- 09:00: Local IT support personnel delivered the missing system memory capture file, explaining it had been prematurely moved to archived files, delaying its discovery by three days. Thanked the IT personnel for their diligence and formally inquired if the team anticipated finding any other information on the corporate system that might affect the ongoing investigation.

- 09:15: The HR representative arrived demanding an update on the spot, stating they were tracking the released employee's location. Maintained a professional demeanor and gently explained that, per established directives, on-the-spot updates could not be provided while the investigation was in active progress.

- 09:30: An email update was dispatched to the investigative manager. The email formally requested that all persons with need-to-know access (including HR) be included in future scheduled email updates to streamline communication. Also requested managerial confirmation that ad-hoc, on-the-spot updates should only be provided directly to supervisors.

Reference (APA 7th ed.)

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